Atakan Topaloğlu

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Education



ETH Zürich | Zürich *September 2025 – Expected Graduation: June 2027*
MSc Electrical Engineering and Information Technology, GPA: 5.49/6.00
IEEE Signal Processing Society Scholarship - **continued**

Koç University | Istanbul *September 2021 – June 2025*
BSc in Electrical and Electronics Engineering, GPA: 3.99/4.00, Department Rank: 2/135
IEEE Signal Processing Society Scholarship **- first recipient from a university in Turkey**

Work & Research Experience

Flexion Robotics | AI Engineering Intern *May 2026 – Present*
• Increased physics simulation **real-time factor from 0.44 to 0.90-1.00** on an RTX 5090 by eliminating unnecessary H2D syncs, reducing GIL contention, and fixing excessive collision-contact generation.
• Accelerated robot visual localization **~5×** by redesigning retrieval and matching, preserving accuracy while reducing localization **GPU memory by 73%.** Cut robot-stack unified memory usage by **47%** by releasing cached GPU allocations after map construction and constructing map histograms slice by slice, which enabled edge deployment.
• Built an **end-to-end lidar-visual-inertial reconstruction pipeline** from raw data to sim exports, outperforming industry baseline in floor planarity and lidar coverage within 5cm, matching 3DGS on SSIM for small/medium sized scenes.

ETH Zurich | Research Assistant, PRS Lab *September 2025 – March 2026*
• Redesigned point-cloud preprocessing by replacing repeated per-instance NumPy queries with a hash-based point-to-object index, reducing runtime from **8 hours to <2 minutes.**
• Diagnosed an I/O bottleneck (<1% GPU utilization), moved data augmentation offline, and added bounding-box rejection before radius-based point filtering, cutting end-to-end training from **12 days to 8 hours.**

Google | Research Collaborator *March 2025 – September 2025*
• Developed OracleGS, achieving **state-of-the-art sparse-view Gaussian Splatting**, filtering generative priors by repurposing a regression model as an uncertainty oracle. Open-sourced the **CUDA kernel** for depth supervision in 3DGS-MCMC.
• Led the project from proposal to publication; first author, **WACV 2026 Oral (top 3%).**

Technical University of Munich | Undergraduate Summer Researcher *July 2024 – September 2024*
• Repurposed a modified neural audio codec for vibrotactile data compression, achieving >30x compression ratios while preserving signal fidelity and supporting real-time CPU inference.

Siemens | Part-Time R&D Working Student *August 2023 – July 2025*
• Developed a CPU-based 3DGS viewer that renders 300k gaussians at 14 FPS, using AoS-to-SoA, SIMD, and octrees with frustum culling to improve cache locality and eliminate rendering of invisible geometry.
• Reduced peak GPU memory in a 3DGS reconstruction pipeline by 83% through color quantization and post-SfM subsampling, enabling processing on a 4 GB NVIDIA Quadro P1000.

Skills

Languages, Systems and Tools: Python, PyTorch, C++20, Docker, Git, CMake, ONNX, SQL
Performance: SIMD, GPU/CPU profiling, Nsight, TensorRT
Languages (Spoken): German (Intermediate), English (TOEFL: 119/120), Turkish (Native)